

Cardiac Output, Blood Flow, and Blood Pressure

Human Physiology - Chapter 14

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Outline

14.1 Cardiac Output

14.2 Blood Volume

14.3 Vascular Resistance to Blood Flow

14.4 Blood Flow to the Heart and Skeletal Muscles

14.5 Blood Flow to the Brain and Skin

14.6 Blood Pressure

14.7 Hypertension, Shock, and Congestive Heart Failure

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Introduction to Cardiac Output

- **Cardiac output** (心输出量): Volume of blood pumped by each ventricle per minute
- Formula: Cardiac output = **Stroke volume** (每搏输出量) × **Heart rate** (心率)
- Normal resting value: 5 L/min
- Can increase up to 25-35 L/min during maximal exercise
- Critical determinant of oxygen delivery to tissues

Autonomic Control of Heart Rate

- Heart rate is controlled by the **autonomic nervous system** (自主神经系统)
- **Sympathetic stimulation** (交感刺激): Increases heart rate (**positive chronotropic effect**, 正性变时作用)
- **Parasympathetic stimulation** (副交感刺激) via **vagus nerve** (迷走神经): Decreases heart rate (**negative chronotropic effect**, 负性变时作用)
- The **SA node** (窦房结) acts as the natural pacemaker

Pacemaker Potential in SA Node

- The **SA node** (窦房结, sinoatrial node) generates spontaneous depolarization called the **pacemaker potential** (起搏器电位)
- Sympathetic nerves release **norepinephrine** (去甲肾上腺素), increasing the rate of depolarization
- Parasympathetic nerves release **acetylcholine** (乙酰胆碱), decreasing the rate of depolarization
- This determines the baseline heart rate



Effects of Autonomic Nerves on the Heart

- Autonomic nerves affect multiple aspects of cardiac function
- Sympathetic activation increases heart rate, conduction velocity through the **AV node** (房室结), and force of contraction
- Parasympathetic activation primarily decreases heart rate and slows AV node conduction
- The table summarizes the comprehensive effects on different cardiac regions



Regulation of Stroke Volume

- **Stroke volume** (每搏输出量): Volume of blood ejected per heartbeat (70-80 mL)
- Regulated by three main factors:
 - **End-diastolic volume (EDV, 舒张末期容积)** - intrinsic control
 - Autonomic nerve activity - extrinsic control
 - **Afterload** (后负荷) - pressure against which heart pumps

Frank-Starling Law and Sympathetic Effects

- The **Frank-Starling law** (弗兰克-斯塔林定律): Increased EDV leads to increased stroke volume
- Sympathetic stimulation produces a **positive inotropic effect** (正性肌力作用)
- With sympathetic stimulation, the heart ejects more blood at any given EDV
- This is shown by the steeper curves when sympathetic nerves are active



The Frank-Starling Mechanism

- Greater stretch of cardiac muscle fibers results in stronger contraction
- This is the intrinsic ability of the heart to adapt to changing venous return
- The mechanism involves optimal overlap of **actin** (肌动蛋白) and **myosin** (肌球蛋白) filaments
- **Preload** (前负荷) refers to the degree of ventricular stretch before contraction



Muscle Length and Contractility

- Cardiac muscle shows a steeper decline in contraction strength with decreased length compared to skeletal muscle
- This emphasizes the importance of the Frank-Starling relationship in cardiac physiology
- **Epinephrine** (肾上腺素) increases contraction strength at any muscle length
- This demonstrates extrinsic control overlaying intrinsic mechanisms



Integrated Regulation of Cardiac Output

- Multiple factors work together to regulate cardiac output
- Stimulatory factors (solid arrows) include sympathetic activity, increased venous return, and increased EDV
- Inhibitory factors (dashed arrows) include parasympathetic activity and decreased venous return
- This integrated control allows fine-tuning of cardiac performance



Blood Distribution in the Circulatory System

- The **venous system** (静脉系统) contains 60-70% of total blood volume
- Veins act as a blood reservoir that can be mobilized when needed
- During exercise, **venoconstriction** (静脉收缩) can shift blood from veins to active circulation
- This is an important mechanism for increasing venous return



Factors Affecting Venous Return

- **Venous return** (静脉回流): Volume of blood returning to the heart per minute
- Key factors include sympathetic **venoconstriction** (静脉收缩), **skeletal muscle pump** (骨骼肌泵), and **respiratory pump** (呼吸泵)
- Negative intrathoracic pressure during inspiration draws blood toward the heart
- Increased venous return leads to increased EDV and thus increased stroke volume via Frank-Starling mechanism



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Body Water Distribution

- Total body water is divided into **intracellular** (细胞内) and **extracellular** (细胞外) compartments
- Extracellular fluid includes **blood plasma** (血浆) and **interstitial fluid** (组织液)
- Plasma comprises about 3.0-3.5 L, while interstitial fluid is 11-13 L
- Water balance involves intake through drinking/food and loss through kidneys, lungs, skin, and GI tract



Starling Forces and Capillary Exchange

- Fluid movement across capillary walls is governed by **Starling forces** (斯塔林力)
- At the arteriolar end, **hydrostatic pressure** (静水压) exceeds **colloid osmotic pressure** (胶体渗透压), causing filtration out
- At the venular end, colloid osmotic pressure exceeds hydrostatic pressure, causing reabsorption
- The balance of these forces determines net fluid movement



Causes of Edema

- **Edema** (水肿): Abnormal accumulation of interstitial fluid
- Can result from increased capillary pressure, decreased plasma proteins, increased capillary permeability, or blocked **lymphatic drainage** (淋巴引流)
- Each mechanism disrupts the normal Starling forces balance
- Examples include **hypertension** (高血压), liver disease, kidney disease, and inflammation



Elephantiasis - Lymphatic Blockage

- **Elephantiasis** (象皮病): Severe edema caused by parasitic infection
- Parasitic larvae block **lymphatic vessels** (淋巴管), preventing fluid return
- Results in massive swelling of affected limbs
- Demonstrates the critical importance of lymphatic drainage in fluid balance



ADH and Blood Volume Control

- **ADH** (抗利尿激素, antidiuretic hormone), also called **vasopressin** (加压素)
- Released by posterior pituitary in response to increased plasma **osmolality** (渗透压) or decreased blood volume
- Increases water reabsorption in kidney collecting ducts
- Provides negative feedback control of blood volume and osmolality



Renin-Angiotensin-Aldosterone System

- Complex hormonal system for regulating blood volume and pressure
- **Renin** (肾素) converts angiotensinogen to **Angiotensin I** (血管紧张素 I)
- **ACE** (血管紧张素转换酶) converts Angiotensin I to **Angiotensin II** (血管紧张素 II)
- Angiotensin II causes **vasoconstriction** (血管收缩) and stimulates **aldosterone** (醛固酮) release



Renin-Angiotensin-Aldosterone System

- Aldosterone increases sodium and water reabsorption in kidneys

Atrial Natriuretic Peptide (ANP)

- **ANP** (心房钠尿肽): Hormone secreted by stretched atrial cells
- Released when blood volume is excessive
- Increases urinary excretion of sodium and water (**natriuresis**, 钠尿)
- Inhibits renin, aldosterone, and ADH - works antagonistically to the renin-angiotensin-aldosterone system
- Provides negative feedback to reduce excessive blood volume



Cardiac Output Distribution at Rest

- At rest, blood flow is distributed based on metabolic needs of organs
- Brain, kidneys, and liver receive disproportionately high blood flow relative to their mass
- Skeletal muscles receive moderate flow at rest
- Distribution changes dramatically during exercise and stress states



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Pressure Difference Drives Blood Flow

- Blood flow is driven by pressure gradients, analogous to electrical current (Ohm's law)
- In systemic circulation, flow is driven from aorta (100 mmHg) to vena cava (0 mmHg)
- The pressure difference (ΔP) is the driving force for blood flow
- Flow = ΔP / Resistance



Vessel Radius and Blood Flow

- **Poiseuille's law** (泊肃叶定律): Flow is proportional to the fourth power of vessel radius
- Doubling the radius increases flow 16-fold; halving the radius decreases flow 16-fold
- This makes vessel radius the most powerful regulator of blood flow
- **Arterioles** (小动脉) are the primary resistance vessels due to their small, adjustable radius



Blood Pressure Through the Circulation

- Blood pressure drops progressively as blood flows through the systemic circulation
- The largest pressure drop occurs in **arterioles** (小动脉) and **capillaries** (毛细血管)
- This is where most of the vascular resistance is located
- By the time blood reaches veins, pressure is very low (10 mmHg or less)



Parallel Organization of Circulation

- Most organs receive blood in **parallel** (并联) rather than **series** (串联) arrangement
- Each organ receives freshly oxygenated blood from the aorta
- Parallel arrangement allows independent regulation of flow to each organ
- Total resistance is less than individual resistances in parallel circuits



Extrinsic Control of Vascular Resistance

- **Extrinsic control** (外在调控):
Regulation by autonomic nervous system and hormones
- Sympathetic nerves cause **vasoconstriction** (血管收缩) via α -adrenergic receptors
- **Epinephrine** (肾上腺素) and **Angiotensin II** (血管紧张素 II) are powerful vasoconstrictors
- Effect varies by organ - brain and heart have minimal extrinsic control, dominated by metabolic autoregulation



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Coronary Circulation and Ischemia

- The heart has very high oxygen demands and relies almost entirely on aerobic metabolism
- **Coronary arteries** (冠状动脉) supply the heart muscle
- Blood flow regulated primarily by **metabolic autoregulation** (代谢性自身调节)
- **Myocardial ischemia** (心肌缺血) occurs when blood flow is inadequate, potentially leading to heart attack



Coronary Artery Bypass Surgery

- Treatment for severe coronary artery blockage
- **Saphenous vein** (隐静脉) segments are commonly used as bypass grafts
- Bypasses blocked coronary arteries to restore blood flow to heart muscle
- Alternative treatments include **angioplasty** (血管成形术) and **stent** (支架) placement



Skeletal Muscle Blood Flow Changes

- Skeletal muscle blood flow varies dramatically between rest and exercise
- At rest: receives 15-20% of cardiac output with moderate flow
- During heavy exercise: can receive up to 80% of cardiac output
- Increase is due to both increased total cardiac output and redistribution of blood flow
- Vascular resistance decreases dramatically in active muscles



Age and Maximum Heart Rate

- Maximum heart rate decreases progressively with age
- Can be estimated by: Maximum HR $\approx 220 - \text{age (in years)}$
- This limits maximum cardiac output in older individuals
- Important consideration for exercise prescription and cardiovascular assessment



Blood Flow Distribution During Exercise

- Dramatic redistribution of **cardiac output** (心輸出量) during exercise
- Total cardiac output increases from 5 L/min to 25 L/min during heavy exercise
- Skeletal muscle flow increases from 15% to 80% of total cardiac output
- Blood flow to muscles increases more than 20-fold
- Brain maintains constant absolute flow; digestive organs and kidneys receive reduced percentage



Cardiovascular Changes During Exercise

- Exercise produces integrated cardiovascular responses
- Increased cardiac output (both heart rate and stroke volume increase)
- Increased **arteriovenous oxygen difference** (动静脉氧差)
- Systolic blood pressure increases; diastolic pressure stable or slightly decreased
- Total peripheral resistance decreases due to **vasodilation** (血管舒张) in muscles



Integrated Cardiovascular Response to Exercise

- Two main adaptations to exercise:
 - (1) Increased cardiac output and total blood flow
 - (2) **Vasodilation** (血管舒张) in exercising muscles with redistribution of blood
- Sympathetic activation increases heart rate, contractility, and causes selective vasoconstriction
- Metabolic vasodilation in active muscles overrides sympathetic vasoconstriction



Integrated Cardiovascular Response to Exercise

- Result: More blood flows to where it's needed most

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Regional Cerebral Blood Flow

- Brain blood flow shows regional variation based on neuronal activity
- Active brain regions receive increased blood flow due to local metabolic demands
- (a) Visual task activates **visual cortex** (视觉皮层) in occipital lobe
- (b) Listening activates **auditory cortex** (听觉皮层) in temporal lobe
- (c) Motor activity activates **motor cortex** (运动皮层)
- (d) Speaking activates multiple cortical areas



Cutaneous Circulation for Thermoregulation

- Skin circulation serves thermoregulation more than nutrition
- **Arteriovenous anastomoses (AVAs, 动静脉吻合)** are unique vascular structures
- AVAs can shunt blood directly from arterioles to venules, bypassing capillaries
- When open, AVAs allow large blood flow for heat dissipation
- When closed, blood flow to skin is minimized to conserve heat



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Effect of Vasoconstriction on Pressure

- **Vasoconstriction** (血管收缩) affects pressure both upstream and downstream
- Constriction increases pressure upstream (arterial side)
- Constriction decreases pressure downstream (venous side)
- This principle explains how arterioles regulate both local blood flow and systemic blood pressure



Cross-Sectional Area and Pressure

- Inverse relationship between total cross-sectional area and blood pressure
- As blood flows from aorta to smaller vessels, total cross-sectional area increases dramatically
- Despite increased area, pressure decreases due to resistance in arterioles
- Capillaries have the largest total cross-sectional area but low pressure
- This allows slow flow for efficient exchange across capillary walls



Baroreceptor Response to Pressure

- **Baroreceptors** (压力感受器) are stretch receptors that detect blood pressure changes
- Located in **carotid sinus** (颈动脉窦) and **aortic arch** (主动脉弓)
- Increased blood pressure → increased stretch → increased action potential frequency
- Signals are sent to cardiovascular control centers in **medulla oblongata** (延髓)
- This provides continuous monitoring of blood pressure



Baroreceptor Reflex Pathway

- Sensory information from baroreceptors travels via cranial nerves to the medulla
- **Cardiovascular control centers** (心血管调节中心) integrate the information
- Output affects both heart and blood vessels via autonomic nerves
- Sympathetic nerves increase heart rate, contractility, and cause vasoconstriction
- Parasympathetic (vagus) nerves decrease heart rate



Negative Feedback Control by Baroreceptors

- **Baroreceptor reflex** (压力感受器反射) provides rapid blood pressure regulation
- Increased pressure → baroreceptors fire more → decreased sympathetic, increased parasympathetic activity → pressure decreases
- Decreased pressure → baroreceptors fire less → increased sympathetic, decreased parasympathetic activity → pressure increases



Negative Feedback Control by Baroreceptors

- Critical for maintaining blood pressure when standing (**orthostatic** (直立性) regulation)

Blood Pressure Measurement Equipment

- Blood pressure measured indirectly using a **sphygmomanometer** (血压计)
- Consists of inflatable cuff and pressure gauge
- Stethoscope used to listen for **Korotkoff sounds** (柯氏音)
- **Auscultatory** (听诊) method is standard clinical technique
- Non-invasive and reliable when performed correctly



Blood Flow and Korotkoff Sounds

- When cuff pressure $>$ systolic pressure: artery completely compressed, no blood flow, no sound
- When cuff pressure between systolic and diastolic: intermittent blood flow, turbulent flow produces Korotkoff sounds
- When cuff pressure $<$ diastolic pressure: continuous blood flow, sounds disappear
- This principle allows indirect measurement of blood pressure



Auscultatory Method of Measurement

- Inflate cuff above **systolic pressure** (收缩压), then slowly deflate
- First Korotkoff sound heard = systolic blood pressure (120 mmHg)
- Sounds continue as cuff pressure decreases
- Last Korotkoff sound heard = **diastolic pressure** (舒张压) (80 mmHg)
- Normal blood pressure: 120/80 mmHg or lower



Five Phases of Korotkoff Sounds

- Phase I: First clear tapping sounds (= systolic pressure)
- Phase II: Sounds become softer, may have swishing quality
- Phase III: Sounds become louder and crisper
- Phase IV: Sounds become muffled
- Phase V: Sounds disappear completely (= diastolic pressure)
- Not all phases heard in all individuals



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Blood Pressure Classification

- **Hypertension** (高血压): Chronically elevated blood pressure
- Normal: $< 120/80$ mmHg
- Prehypertension: $120\text{-}139/80\text{-}89$ mmHg
- Stage 1 hypertension: $140\text{-}159/90\text{-}99$ mmHg
- Stage 2 hypertension: $\geq 160/\geq 100$ mmHg
- Higher pressures increase risk of stroke, heart attack, and organ damage



Secondary Hypertension Causes

- **Secondary hypertension** (继发性高血压): Elevated blood pressure with identifiable cause (5-10% of cases)
- Renal causes: kidney disease, renal artery stenosis
- Endocrine causes:
pheochromocytoma (嗜铬细胞瘤),
Cushing syndrome,
hyperaldosteronism
- Vascular causes: coarctation of aorta
- Treating the underlying cause can resolve the hypertension



Antihypertensive Drug Mechanisms

- Multiple drug classes available to treat hypertension
- **Diuretics** (利尿剂): Reduce blood volume
- **ACE inhibitors** (ACE 抑制剂) and **ARBs** (血管紧张素受体阻滞剂): Block angiotensin II effects
- **Beta-blockers** (β 受体阻滞剂): Reduce heart rate and contractility
- **Calcium channel blockers** (钙通道阻滞剂): Cause vasodilation
- Often combination therapy is most effective



Clinical Signs of Shock

- **Shock** (休克): Life-threatening condition of inadequate tissue perfusion
- Signs include rapid heart rate (**tachycardia**, 心动过速), weak pulse, low blood pressure (**hypotension**, 低血压)
- Cold and clammy skin, reduced urine output (**oliguria**, 少尿)
- **Metabolic acidosis** (代谢性酸中毒) develops as tissues become hypoxic
- Requires emergency treatment



Compensatory Reflexes in Shock

- Body activates multiple compensatory mechanisms to maintain blood pressure during shock
- **Baroreceptor reflex** (压力感受器反射): Increases sympathetic activity
- **Chemoreceptor reflex** (化学感受器反射): Responds to hypoxia and acidosis
- Hormonal responses: Release of epinephrine and ADH
- These reflexes attempt to maintain perfusion to vital organs (brain, heart)

